Quick sort

- select pivot
- rearrange list so first s are smaller and n-s are larger
- exchange pivot with last element in first subarray
- sort the two recursively

Best case: $\Theta(n \log n)$

Worst: $\Theta(n^2)$

Average $\Theta(n \log n)$

Hoar's Partitioning Algorithm:

- scan subarray from both ends, comparing subarray's to pivot
- left-to-right denoted by i starts with second element
- scan stops over elements < pivot and stops when ≥ pivot
- right to left denoted by j starts with last of subarray
- scan stops over elements > pivot and stops when ≤ pivot
- after scans stop:
  1) if indices i and j haven't crossed, exchange A[i] and A[j], resume incrementing i and decrementing j
  2) if indices crossed over, partition the subarray after exchanging pivot and A[j]
  3) if indices stop and point to same element, s = i = j

Quicksort example: 5 3 1 9 8 2 4 7

0	1	2	3	4	5	6	7
5	3	1	9	8	2	4	7
5	3	1	4	8	2	6	7
5	3	1	4	8	2	4	7
5	3	1	4	8	4	9	7
5	3	1	4	2	8	9	7
5	3	1	4	2	8	4	7
5	3	1	4	5	8	4	7

↳ 2 3 1 4 8 9 7

2 3 1 4 8 9 7

2 3 1 4 8 9 7

2 1 3 4 8 9 7

2 1 3 4 7 8 9

1 2 3 4 9

↳ 3 4

3 4 Result: 1 2 3 4 5 7 8 9

Example: 3 4 0 1 2 5

0	1	2	3	4	5
3	4	0	1	2	5
3	4	0	1	2	5
3	4	0	1	2	5

5 4 0 1 2

5 4 0 1 2

5 4 0 1 2

5 4 0 1 2 Result: 0 1 2 3 4 5

4 5 0 1 2

↳ 0 1 4 5

0 1 4

↳ 0 1 2